

Simulation of $SU(N)$ Gauge Theories via Equivariant GNNs and Normalizing Flows

Updated Mathematical and Physical Technical Report

1 Introduction

This report details a computational framework for simulating lattice gauge theories using Graph Neural Networks (GNNs). The code implements a discrete version of the Standard Model's gauge group structure: $G = SU(3) \times SU(2) \times U(1)$. By utilizing gauge-equivariant message passing, the model preserves the fundamental symmetries of particle physics while leveraging Stochastic Gradient Langevin Dynamics (SGLD) for quantum ensemble sampling.

2 Mathematical Framework

2.1 Gauge Group Generators

The model defines the Lie algebra generators for the three components of the Standard Model. These matrices form the basis for the gauge fields A_μ :

- **SU(2) (Weak Interaction):** Represented by the three Pauli matrices σ_a .
- **SU(3) (Strong Interaction):** Represented by the eight Gell-Mann matrices λ_a .
- **U(1) (Electromagnetism/Hypercharge):** Represented by complex phase rotations.

The gauge links U_{ij} are generated via the exponential map from the algebra to the group:

$$U_{ij} = \exp\left(i \sum_a \alpha_a T^a\right)$$

where T^a are the generators and α_a are the learned parameters (gauge potentials).

2.2 Complex Activations

The core linear transformation in the `ComplexLinear` layer has been upgraded from a purely linear to an affine transformation by adding a complex bias $b = b_{\text{re}} + ib_{\text{im}}$:

$$f(z) = Wz + b$$

This allows the model to learn global field offsets, which is critical for identifying the specific vacuum state during symmetry breaking.

2.3 Learnable ModReLU Activation

To handle complex-valued matter fields $\varphi \in \mathbb{C}^n$, the code implements the `mod_relu` activation function:

$$\text{mod_relu}(z) = \begin{cases} (|z| + b) \frac{z}{|z|} & \text{if } |z| + b > 0 \\ 0 & \text{otherwise} \end{cases}$$

By making the bias b a learnable parameter per channel, the network can dynamically adjust the “dead zone” magnitude where fluctuations are suppressed.

3 Physics of the Architecture

3.1 Parallel Transport and Covariant Convolution

In a gauge theory, comparing field values at different points (φ_i and φ_j) is physically meaningless without a connection. The SUN_GaugeConv layer implements **Parallel Transport**.

When a message is passed from node j to node i , it is multiplied by the gauge link U_{ij} :

$$\varphi'_i = \sum_{j \in \mathcal{N}(i)} U_{ij} \varphi_j$$

This ensures that the output is **Gauge Equivariant**. If we perform a local gauge transformation $\varphi_i \rightarrow g_i \varphi_i$, the link transforms as $U_{ij} \rightarrow g_i U_{ij} g_j^\dagger$, maintaining the consistency of the field interactions.

3.2 The Action (Loss Function)

The training objective is the minimization of the Euclidean Action $S[\varphi, U]$, consisting of three primary terms:

1. **Potential Term (S_V):** Implements the “Mexican Hat” potential for Spontaneous Symmetry Breaking (SSB):

$$S_V = \lambda \left(\sum_i |\varphi_i|^2 - v^2 \right)^2$$

This drives the field to a Vacuum Expectation Value (VEV) of v .

2. **Kinetic Term (S_K):** The discrete covariant derivative:

$$S_K = \kappa \sum_{\langle i, j \rangle} |\varphi_i - U_{ij} \varphi_j|^2$$

This penalizes sharp variations in the field that are not aligned with the gauge field.

3. **Wilson Plaquette Action (S_W):** Represents the pure gauge field energy:

$$S_W = \beta \sum_{\square} \left(1 - \frac{1}{N} \text{Re Tr}[U_1 U_2 U_3 U_4] \right)$$

where the sum is over all elementary squares (plaquettes) on the lattice.

4 Sampling and Optimization

4.1 Burn-in and Thermalization

The simulation uses the **Adam** optimizer for an initial “burn-in” phase. This is physically equivalent to **Thermalization**, where the system is driven from a random high-energy state toward the local minimum of the action (the classical configuration).

4.2 Stochastic Gradient Langevin Dynamics (SGLD)

After burn-in, the code transitions to SGLD. SGLD adds Gaussian noise to the gradients:

$$\theta_{t+1} = \theta_t - \varepsilon \nabla S(\theta_t) + \sqrt{2\varepsilon T} \eta_t$$

where $\eta_t \sim \mathcal{N}(0, 1)$. This allows the model to sample from the Boltzmann distribution $P(\varphi) \propto \exp(-\frac{S}{T})$, effectively simulating **Quantum Fluctuations** around the vacuum.

4.3 Gauge-Equivariant Convolution with Normalization

The `SUN_GaugeConv` layer performs parallel transport of fields across edges. To prevent runaway weight scaling, a normalization step is applied after the linear transformation:

$$\varphi'_i = \frac{\sum U_{ij} \varphi_j}{\langle |\varphi| \rangle + \varepsilon}$$

This ensures that the GNN learns the spatial correlations of the field without being biased by its absolute magnitude.

4.4 Jackknife Error Estimation

For the mass gap calculation, the model employs **Jackknife Resampling**. This provides a robust estimate of statistical errors for the effective mass $m_{\text{eff}}(t)$:

$$\sigma_m(t) = \sqrt{\frac{N-1}{N} \sum_{i=1}^N (\hat{m}_i(t) - \langle m \rangle(t))^2}$$

where $\hat{m}_i$ is the mass calculated from the i -th jackknife block. This method is essential for identifying stable mass plateaus amidst quantum noise.

5 Physical Observables

5.1 Vacuum Expectation Value (VEV)

The model tracks the average magnitude of the Higgs-like field:

$$v = \langle |\varphi| \rangle$$

A stable, non-zero VEV indicates that the $SU(2)$ symmetry has been successfully broken.

5.2 Mass Gap Calculation

The mass of the scalar particle is extracted from the exponential decay of the two-point correlation function $C(t)$:

$$C(t) = \langle \varphi(0) \varphi(t) \rangle \sim \exp(-mt)$$

The code utilizes the `fsolve` method to find the effective mass m using the cosh-ratio method, which accounts for the periodic boundary conditions on a finite lattice:

$$\frac{C(t)}{C(t+1)} = \frac{\cosh(m(t - \frac{L}{2}))}{\cosh(m(t+1 - \frac{L}{2}))}$$

6 Conclusion

The provided implementation bridges the gap between Deep Learning and Lattice Field Theory. By embedding the $SU(N)$ group structure directly into the GNN architecture, it provides a robust platform for studying phase transitions, symmetry breaking, and particle masses in a gauge-invariant manner.